

Simulation of Heatwave event using WRF Model across Uttar Pradesh, India

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Abstract— Numerical weather prediction (NWP) models are being used to predict and minimise the effects of the increasing number of extreme weather events occurring globally. Applying NWP models necessitates choosing the right physics parametrisation choices in order to closely describe atmospheric processes. The simulations of this work in to determine episodic Heat Wave (HW) events from 20th - 24th May 2023 over Uttar Pradesh (UP) using Weather Research and Forecasting (WRF) model, India with varying horizontal resolutions of 12 km for the whole of India (d01), and 4 km for the UP (d02) domain. In this study, the maximum temperature (Tmax) is analysed using WRF model during HW occurrences and this result is validated with fifth generation of the European reanalysis (ERA5). The experiment produces the temperature on this scale on the desired domain using Yonsei University Schemes (Planetary Boundary layer). The model shows uncertainty in the Tmax for the regions, indicating that complicated orography and location affect regional temperature. Furthermore, the characteristics associated to HWs like Tmax, relative humidity (RH) distribution is analysed to comprehend the dynamics of increasing HW intensity. Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Correlation Coefficient (CC) for Tmax and relative humidity analysis are acquired in order to verify the findings with existing literature.

Keywords— Heatwave, WRF model, PBL sensitivity ERA5, Uttar Pradesh.

I. INTRODUCTION

Air temperature is a basic meteorological variable that effects the thermal energy balance between the atmosphere and the land (Baldocchi and Ma, 2013). Temperature fluctuations directly affect human health, economy, farming, water supplies, and the ecosystems in addition to upsetting the atmospheric condition and causing dangerous weather and climate events. The mean temperature for the 20th century was 13.9°C (+57.0°F), whereas the hottest year since global records began in 1850 was 1.18°C (+2.12°F). In Asia, the mean from 1910–2023 was +0.18°C (+0.32°F) each decade; however, the trend from 1982–2023 is more than double as high as the longer-term trend (+0.41°C/74°F). Indian subcontinent, the second-largest country in Asia, has shown

a rise in temperature during the summer from 1982–2018 at a mean of 0.29°C in every ten years (Sanjay et al., 2020; Karrevula et al., 2023). According to IMD, the mean temperature of previous 15 years (2007–2021) raised more than 70% from the year 1901 (IMD, 2016). According to the previous study, potential causes of rising air temperature in land-use land-cover (LULC) are: different anthropogenic activities, raising Aerosol Optical Density (AOD), Industrialization and emission of CO₂ gas (IPCC, 2019). HWs, which is generally arise in May–June till the starting of the summer season, occur in a several locations in India, especially in the eastern, central, and northwest regions (Rohini et al., 2016; Mandal et al., 2019). Ray et al., (2021) verified that during 1970–2019, 706 HWs events caused 17,362 deaths in India. The mortality rate is 62.2% higher during the period 2000–2019 as compared to 1980–1999. In India, event of HWs and also the duration of HWs raise by *25%, during 1985–2009 (Mazdiyasi et al., 2017). Similarly in India, the air temperature is raising by 0.6 °C in each decade (Bhattacharya et al., 2023).

Temperature, wind and moisture are significantly influenced by the earth's surface because of mass transfer of turbulent air in the lower atmosphere. The planetary boundary layer (PBL) schemes are an important component of the HWs for forecasting improvement. During HWs, temperature elevation, low wind and a rapid rate of heat circulation is observed in a specific PBL (Wu et al., 2019). In order to increase the temperature across an area that could result in the HW situation in a particular PBL. Therefore, the atmospheric air temperature fluctuation with in a particular PBL plays a crucial role in modelling to improve the accuracy of predictions, specifically the rapidly expansion of heat in UP which may minimize the effects of HWs along with death rate. The current study is based on a mesoscale dynamical model for evaluating and forecasting short-range weather system, the WRF model over the region of UP (India).

WRF model performance is dependent on domain size, resolution, and initial and boundary circumstances, parameterization of physical schemes, and modification of

physical processes. To evaluate the air temperature over individual areas, sensitivity experiments using separate and combined parameterization of physical techniques are used (Mohan and Bhati, 2011; Kadaverugu et al., 2021).

The primary scientific goal of this research is to evaluate the WRF model using the PBL scheme in predicting HWs across the UP at a 4 km resolution. The outcomes of this study may be useful to determine the WRF configuration and to appreciate the PBL scheme's role, as well as the improving prediction abilities for model developers and forecasters. Section 2 describes the study area, data and evolution of methodology used, result and discussion section in terms of statistical characteristics is defined in section 3, section 4 describes the silent of the work.

II. DATA AND METHODOLOGY

A. Study Area

UP is one of the major states of India, area-wise fourth in central north east region (23.5 – 30.5 N and 77 – 84.7 E) shown in Fig.1. It is the first largest state in terms of population. According to the 2023 report the population of UP is 23,46,92,000. The monthly average temperature of UP remains between 20.5 – 38 °C while, the yearly average temperature remains 31 °C. However, May is the warmest month, with daily maximum temperature 30 – 42 °C. UP experiences more than four HWs annually between April to June, which makes this region important to study.

B. Data Used

ERA5 is a fifth-generation reanalysis data provided by ECMWF which provides the worldwide data of climate and weather. The ERA5 datasets provides the C3S Climate Data Store (CDS) at 0.25°. In addition the NCEP–GFS Final Analysis (FNL) data which is available at 6 hour intervals and at 0.25° data is simulated with ERA5 data.

C. Methodology

Model configuration

The role of PBL in modelling the HW across the study's region has been evaluate using the high-resolution WRF model version of 4.3 (Skamarock et al., 2019). The two nested domains in the model's setup have horizontal resolutions of 12 km and 4 km: Domain – 01 (4.8 – 44.5N; 55 – 112E), Domain – 02 (21 – 32N; 74.5 – 85E). Domain – 01 covers the entire India, Domain-02 in the study area, it has resolution of 4km and covers UP. TABLE I. and Fig. 1. present more information on the domain setup and model configure. In this study, the PBL schemes at Yonsei University (YSU) to determine their ability in simulations the HW case over UP.

WRF-SM6

Heatwave use the microphysics scheme for WSM6 (Hong and Lim, 2006) predicts six hydrometer classifications: Snow, water vapor, rain, cloud ice, cloud water, and graupel. It is inclusion of mixed-phase and ice developments, this scheme enhances the illustration of cloud microphysics processes, enabling more accurate simulation of cloud formation, precipitation, and melting. WSM6 can more properly handle large rainfall events, convective storms, and deep convection than simpler schemes because it explicitly models graupel, which is frequently linked to intense convective precipitation. It is used to simulate heatwaves by simulating cloud microphysical processes that impact radiation and surface energy balance.

Grell 3D Ensemble

Heatwave use the Grell 3D Ensemble Scheme is Cumulus Scheme (Grell, 1993) is a cumulus parametrisation to model convective phenomena at sub-grid scale that affect the vertical diffusion of moisture and heat in the WRF model. The scheme takes into relationship convection's horizontal and vertical interactions, enabling cloud development to be impacted by the grid environment in all three dimensions. It enhances the simulation of moisture and energy exchange between atmosphere and the surface by resolving entrainment – detrainment processes, updrafts, and downdrafts. In heatwave simulations, it helps represent the suppression of convection and the intensification of surface heating under stable atmospheric conditions.

RRTM Longwave

Heatwave use the RRTM longwave radiation scheme (Mlawer, 1997) is crucial for heatwave simulations since it offers a thorough depiction of air absorption and longwave radiative transmission. The impacts of water vapour, temperature and greenhouse gases on longwave cooling rates are all relationship by RRTM. The equilibrium between atmospheric trapping of longwave radiation and surface thermal emission, especially in dry and clear skies, has a significant impact on heatwaves.

Dudhia Shortwave

Heatwave use the Shortwave radiation of Dudhia scheme (Dudhia, 1989) since it offers a simple and effective representation of incoming solar radiation. During heatwaves, strong shortwave radiation is a major driver of daytime surface heating, especially under clear-sky conditions. The Dudhia scheme accounts for atmospheric scattering and absorption, helping to model the quantity of solar energy reaching the land surface. This directly influences surface temperature, boundary-layer growth, and heatwave intensity. Although computationally light, the scheme captures essential shortwave processes that affect daytime warming and persistence of heatwaves. Its simplicity and stability make it suitable for long-term regional climate and heatwave studies.

Yonsei University Scheme (YSU)

Heatwave use the YSU scheme was the planetary boundary layer scheme (Hong, 2006) was modified from the vertical diffusion scheme to develop the MRF PBL scheme. For free atmospheric diffusion, the K-approach is used as the local dispersion method at the mixed layer. It is a first-order nonlocal closure approach for atmospheric turbulence modelling. It uses a turbulence diffusion equation with diffusion of eddy coefficient and a countergradient correction term to calculate prognostic flow, components of the mixed layer. An explicit parameterisation of entrainment is incorporated into the YSU scheme by adding a factor to the turbulence diffusion equation. The method estimates the PBL height by computing Richardson number (Rib) values from the surface.

MM5

Heatwave use the MM5 scheme (Jimenez et al., 2012) is used in WRF to represents the transfer of heat, momentum, and moisture between the atmosphere and the Earth's surface. The basis of it is the theory of Monin – Obukhov similarity, which accounts for atmospheric stability and surface roughness. Due to dry soil conditions, this scheme aids in simulations strong practical heat fluxes and decreased latent heat instability.

Experimental Design:

WRF Configuration	Description
Resolution	12Km (d01), 4Km (d02)
Map projection	Mercator
Micro-Physics	WRF-SM6 (Hong and Lim, 2006)
Cumulus-Physics	Grell 3D Ensemble Scheme (Grell, 1993), No cumulus for d02
Radiative longwave Physics	RRTM (Mlawer, 1997)
Radiative shortwave Physics	Dudhia (Dudhia, 1989)
Planetary Boundary Layer Physics	YSU (Hong, 2006)
Surface layer	MM5 Schemes (Jimenez et al. 2012)
Land surface	Noah (Tewari, 2004)
Num-land Category	21 (USGS Land Use)

TABLE I. Details about the model configuration.

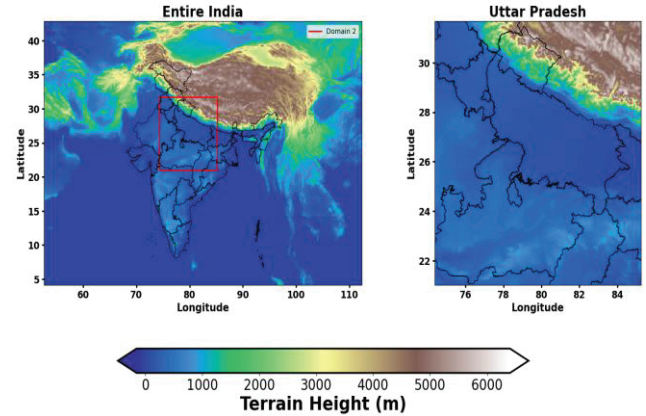


Fig. 1. The WRF model use HW simulations domain. The shaded colour represented by topography from sea level measured (m). UP location represented by red square in the left side. The second domain figure in the WRF model is shown on the right.

Evaluation of Methodology

Several statistical studies, such as mean absolute error (MAE), root mean square error (RMSE), and spatial correlation coefficient (CC), are calculated to evaluate the model's effectiveness. Numerous studies are extensive used for basic statistical methods to assess the model performance (Patil and Pradeep, 2016; Dodla et al., 2017; Jain et al., 2022).

These are equations (1), (2), and (3):

$$MAE = \frac{1}{N} \sum_{j=1}^N |(M_j - O_j)| \quad (1)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{j=1}^N (M_j - O_j)^2} \quad (2)$$

$$CC = \frac{\sum_{j=1}^N ((O_j - \bar{O})(M_j - \bar{M}))}{\sqrt{\sum_{j=1}^N (O_j - \bar{O})^2 \sum_{j=1}^N (M_j - \bar{M})^2}} \quad (3)$$

When M_j and O_j represented the model predicted and observed values respectively. A model that performs better is shown by less MAE, RMSE, and CC. We calculated the HW case forecasts at the Tmax and RH in addition to conducting the statistical analysis.

III. RESULTS AND DISCUSSION

The spatial distribution of ERA5 and evaluated temperature at 2m height (T2m) and relative humidity at 2m height from the PBL (YSU) schemes for the HW event occurrent are represented (Fig. 2.) T2m observed ranges between 35–44°C during HW event over UP (in Fig. 2.). In event (20th–24th May 2023), YSU scheme predicated well, it has simulated maximum temperature over the central portion of the UP; It is most caused by the density of more populated areas above that location. The temperature distribution of the event is represented spatially by its PBL scheme.

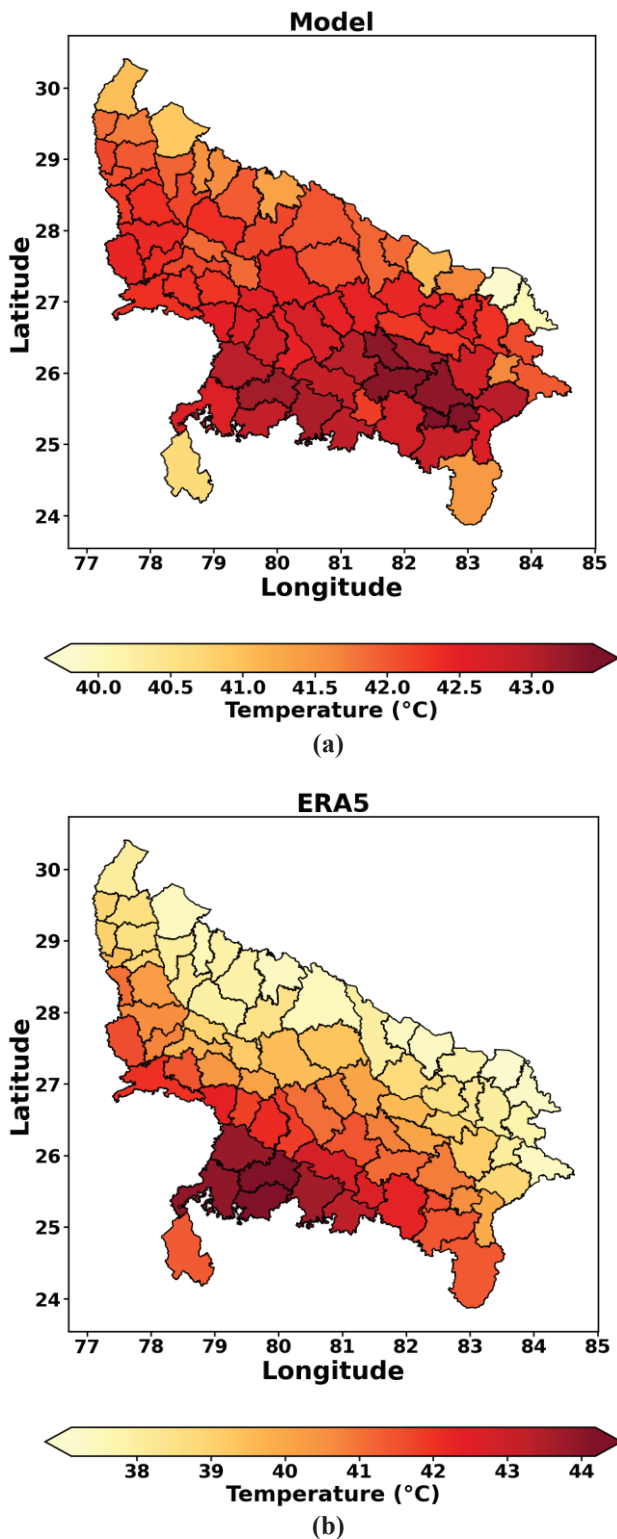


Fig. 2. Spatial distribution of temperature (°C) of (a) WRF model and (b) ERA5, at a height of two meters (T2m) across UP.

The regional distribution of temperature has been analyzed, and as shown to the results that the YSU simulated T2m has succeeded well with ERA5, with the MAE = 2.48 °C, RMSE = 2.95 °C, CC = 0.98. This could be due to increased entrainment at the PBL's top, which increases vertical mixing and reasons of the PBL to become drier and hotter. The prediction for the temperature in the morning is critical since

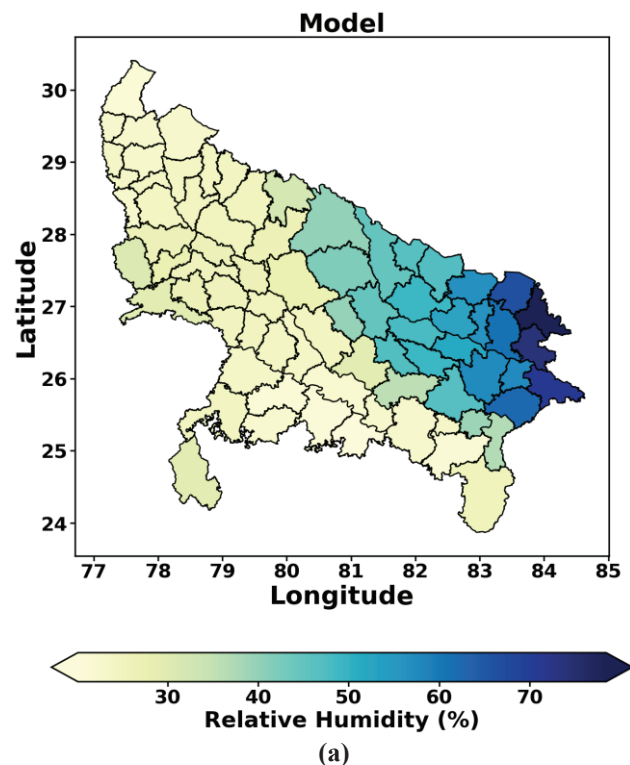
it influences sleeping hours and circadian rhythms on HW days.

The circadian rhythm and sleeping hours during HW days are affected by the early morning temperature forecast, creation it crucial. Disrupted nocturnal, sleep affects every day activities and is associated with numerous negative health consequences, such as obesity, poor life quality, and even mortality (Okamoto – Mizuno and Mizuno, 2012). Examining T2m's peak hours in the model is intriguing (Fig. 2.). The research reveals that the model's T2m peak hours and magnitude closely match the PBL scheme's observations. The YSU scheme generally exhibits the statistical errors (MAE = 2.48 °C, RMSE = 2.95 °C, CC = 0.98 respectively).

Relative Humidity at 2m

Another important factor during a HW is relative humidity (RH), which affects not only the HW's susceptibility but also the comfort of the fauna and Cora. The performance of the PBL system has been calculated by spatial and temporal valuations of RH at 2m for the particular HW event.

When the spatial distribution of RH2 was related to the ERA5 reanalysis data, the YSU scheme was exposed. The YSU scheme for the event (MAE = 24.44 %, RMSE = 25.14 % and CC = 0.89) well illustrates the diurnal variations of RH. However, the model's measured performance is observed in depicting the temporal variability of RH for events over UP.



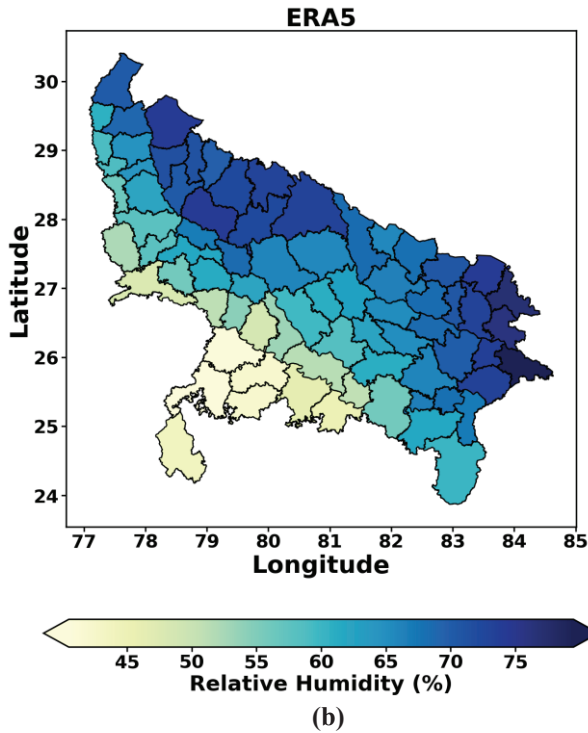


Fig. 3. The spatial distribution of relative humidity (%) (a) WRF model and (b) ERA5, at a height of two meters (T2m) across UP.

Forecast hours:

The forecast of air temperature (T2m) and relative humidity (RH2) of the 2-m height during the 120-hour forecast period was evaluated against ERA5 to compute the temporal model's performance over UP. During the initial forecast hours, high RMSE values and low correlation coefficients for T2m represent that the model performs poorly in replicating near-surface temperature variations relative to ERA5, suggesting deficiencies in the representation boundary-layer processes and of surface energy balance. The forecasted time period increases, a gradual rise in RMSE accompanied by a reduction in correlation suggests growing uncertainty associated with error accumulation and limitations in simulating land-atmosphere interactions. The degradation of skill at longer forecast hours is for RH than for T2m, highlighting the greater sensitivity of moisture variables to surface fluxes, soil moisture initialization, and turbulent mixing. However, the overall consistency between the model and ERA5 during the short- to medium-range forecast demonstrates that the model provides valuable guidance for heatwave diagnostics and humidity-related thermal stress assessments at sub-seasonal time scales.

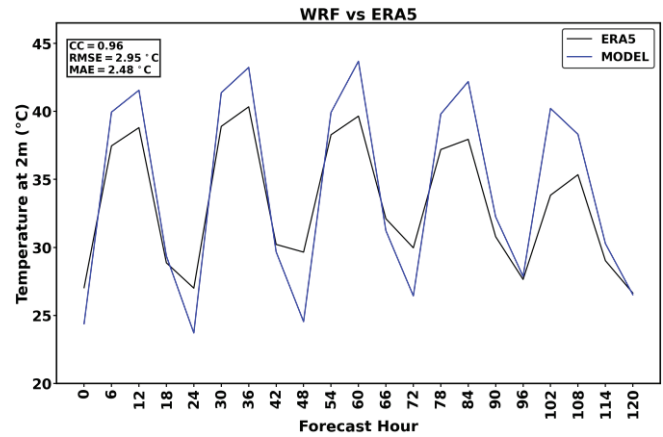


Fig. 4.a. The area-averaged temperature (°C) time series at a 2m height across UP. Black and blue, respectively, represent the ERA5 and model data. The MAE, RMSE and CC for each scheme in relation to the ERA5 and model data represented numerically.

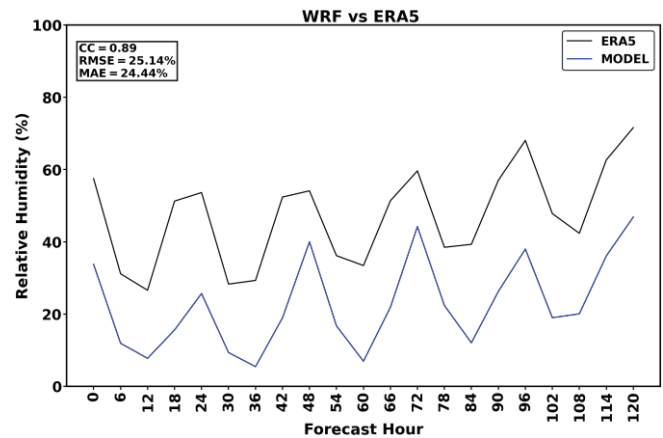


Fig. 4.b. The area-averaged relative humidity (%) time series at a 2m height across UP. Black and blue, respectively, represent the ERA5 and model data. The MAE, RMSE and CC for each scheme in relation to the ERA5 and model data represented numerically.

IV. SUMMARY AND CONCLUSION

Two explanations are the main focus of the present study. First, whether the PBL parameterization scheme and related mechanism are appropriate t2m. The second is the time-based and spatial variability of surface energy atmospheric circulations and relative humidity during HWs time. For HW over the UP, using the ERA5 reanalysis datasets, the model simulated outputs have been validated, and the performance of the YSU PBL scheme using a high-resolution WRF model. The spatial fluctuation of T2m during the midday captured by the YSU signify as T2m's early morning approximately. T2m simulations with warm bias are made worse by YSU capacity to highlight the daytime variability of T2m, but permitting local mixing and increased entrainment at the top. The YSU scheme is not good at illustrating the time-based change of relative humidity and is proficient of causation a more accurate relative humidity pattern. Air temperature and other variables that are strongly correlated with the incidence of HWs can be predicted using the YSU scheme but not for relative humidity. During the HW period UP is warming and the overestimation of the PBL scheme due to changes in atmospheric circulations and land-atmosphere interactions.

Overall, this study suggests understanding information about the YSU PBL scheme in WRF to improve HW prediction and its effects. which, in HWs areas like UP, India, and elsewhere, might be helpful in a variety of real-world applications affecting to policy creation, public healthiness, city planning, climate change edition, and deforestation.

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